

Sathvik Ram Lanka

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EDUCATION

University of Illinois Urbana-Champaign

Incoming Aug 2027

Master of Computer Science

University of Illinois Urbana-Champaign

Expected Graduation: May 2027

B.S. in Computer Science and Statistics, Minor in Mathematics

GPA: 3.6/4.0

- **Relevant Coursework:** Algorithms, Systems Programming, Database Systems, Data Structures, Languages & Automata, Machine Organization, Programming Languages, Distributed Systems, Applied Machine Learning, Statistical Modeling, Statistical Data Management

TECHNICAL SKILLS

Languages: C/C++, Python, SQL, C#, Java, Go, Rust, HTML, CSS, JavaScript, TypeScript, Haskell

Frameworks: React, Node.js, Express, Flask, Tailwind CSS, Three.js, Unity, CUDA

Databases & Infrastructure: PostgreSQL, Redis, MongoDB, MySQL, Docker, Git, AWS, GCP, Linux, GDB

Libraries: Hugging Face, Scikit-Learn, Pandas, Streamlit, gRPC, Protobuf, Vite

EXPERIENCE

ATLAS UIUC

Champaign, IL

Software Developer Intern

August 2025 – December 2025

- Shipped Quad Games, a real-time multiplayer interactive system in C# handling 500+ concurrent users during live campus events, coordinating large codebase changes and weekly live deployments.
- Reduced real-time simulation latency by achieving consistent 90Hz frame delivery for UIUC's PointVR research team, by debugging floating-point precision errors and optimizing the physics computation pipeline across the Unity codebase.
- Prevented production data loss incident and restored access to 2,000+ lines of project state within 2 hours by recovering fragmented Unity project files and re-establishing version control discipline across the team.

OLI Gardens, Patriotic Gardens

Aurora, IL

Software Engineer Intern

May 2025 – August 2025

- Delivered secure, role-based content management across 3 user tiers, as measured by shipping 15+ production RESTful API endpoints, by architecting a full-stack MERN platform in TypeScript, Node.js, and MongoDB.
- Implemented a role-based access control system protecting 50+ admin operations, measured by zero unauthorized access incidents post-security audit, by building JWT authentication and bcrypt hashing with granular permission scoping.
- Enabled recurring and one-time donation processing in production, as measured by successful integration into the platform's revenue infrastructure, by implementing Stripe payment gateways with tier logic and webhook handlers.

PROJECTS

Distributed Configuration Server | Go, Raft Consensus, gRPC, PostgreSQL, Redis, React, TypeScript, Docker

[GitHub](#)

- Architected a Raft-based distributed configuration server with atomic consistency and automatic leader election, validating failover and replication under node failures via persistent PostgreSQL log replication and snapshotting.
- Engineered dual-path Redis reads for μ s retrieval, eliminating database round-trips while enforcing strict consistency via Raft log ordering and write-through cache invalidation.
- Built real-time WebSocket config propagation via WebSocket streaming for zero-downtime updates; Utilized Docker Compose suite validating leader election, config replication, and Byzantine failure benchmarks.

Rigveda Semantic Search Engine | Python, Hugging Face, Scikit-Learn, Pandas, Streamlit

[GitHub](#)

- Built a high-performance retrieval backend supporting semantic search across 10,000+ documents by engineering a Pandas/Parquet data pipeline that parallelizes Hugging Face transformer embeddings into a queryable vector index.
- Delivered a real-time search interface supporting concurrent user searches with sub-500ms response times, as measured by 95% semantic relevance accuracy via Scikit-Learn's sparse matrix operations and in-memory caching.
- Enabled intuitive exploration of semantic relationships across the corpus, as measured by real-time visual navigation of query results, by building an interactive vector space map in Streamlit that plots embedding similarity spatially.

CUDA Mandelbrot Set | C++, CUDA, GPU Acceleration

[GitHub](#)

- Developed a GPU-accelerated Mandelbrot set generator parallelizing the escape-time algorithm across 1,024+ CUDA threads, delivering 4K UHD fractal rendering performance, measured by achieving 8x speedup over optimized CPU implementation while maintaining sub-second frame render times.
- Optimized critical memory and disk I/O bottlenecks by replacing standard ASCII file streams with direct binary memory writing via `reinterpret_cast` and memory-mapped file I/O, reducing 4K image save times by over 90%.
- Engineered a 2x2 supersampling anti-aliasing (SSAA) architecture and continuous logarithmic color smoothing directly within the CUDA kernel, eliminating discrete banding artifacts and generating mathematically seamless gradient transitions at sub-pixel precision.